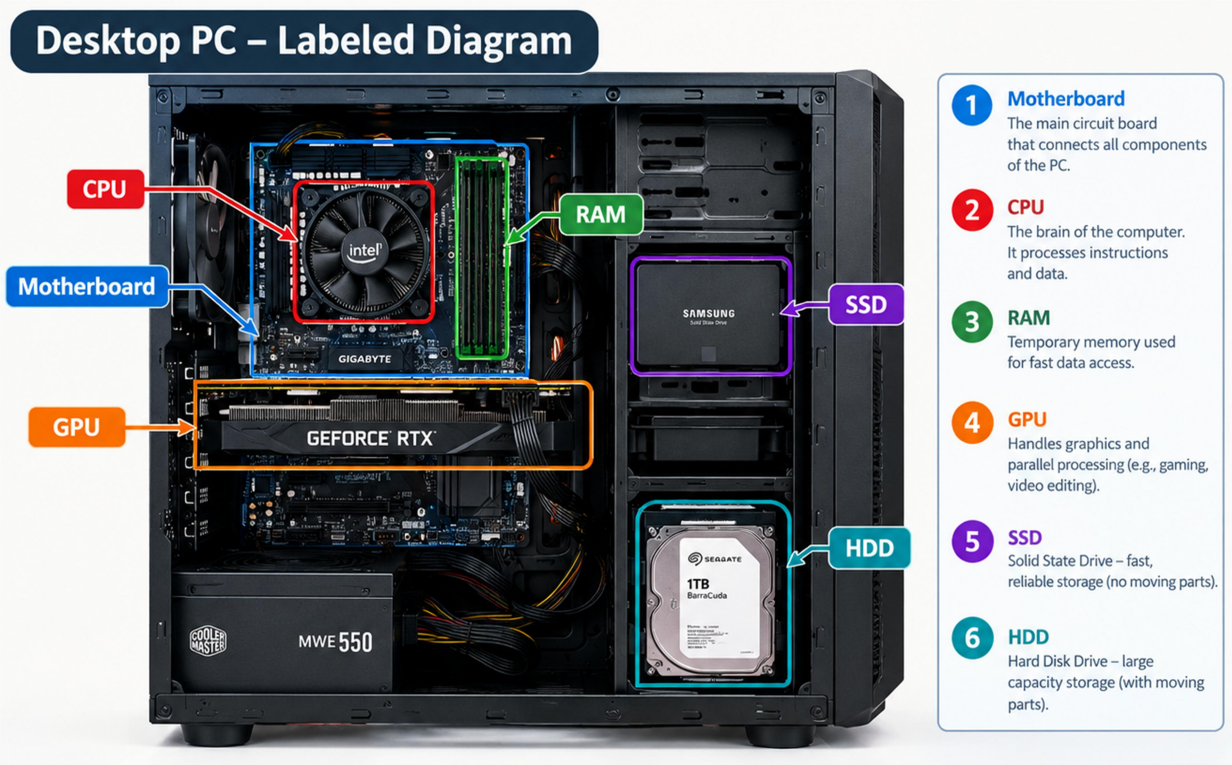
★TOPIC:- HARDWARE

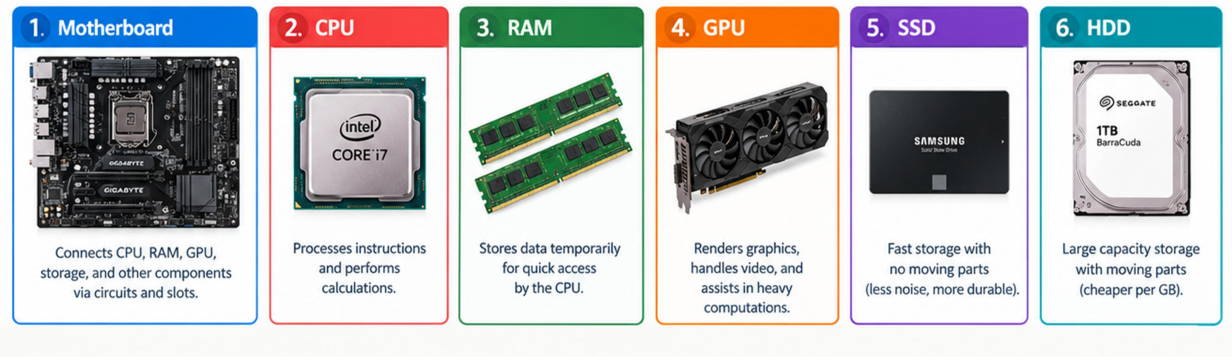
Module_1-Day_1:

❖ Tasks:

1. **Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.**

⇨





2. **Create a compari9son table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).**

⇨

❖ **Difference Between HDD and SSD:**

Feature	HDD	SSD
Full Form	Hard Disk Drive	Solid State Drive
Speed	Slower	Much faster
Durability	Less shock-resistant because it has moving parts	More shock-resistant because it has no moving parts
Noise	Makes some noise	Silent operation
Cost	Cheaper	More expensive
Storage Capacity	Usually offers more storage at a lower cost	Usually costs more for the same capacity
Typical Use	File storage, backups, and budget computers	Laptops, gaming PCs, and operating systems

3. **Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.**

⇨

❖ **CPU (Central Processing Unit):** The CPU processes the game's instructions and controls game logic, player movements, and enemy actions. It helps the game run properly and responds quickly to the player's actions.

❖ **RAM (Random Access Memory):** RAM temporarily stores the game data, maps, and other information needed while playing. Sufficient RAM helps the game run smoothly and reduces lag.

❖ **GPU (Graphics Processing Unit):** The GPU renders the game's graphics, characters, environments, and visual effects. A powerful GPU provides better image quality, higher frame rates, and a smoother gaming experience.

4. **Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.**

⇨

- **GPU (Graphics Processing Unit):** Upgrade to a powerful graphics card to improve graphics quality, frame rates, and smooth gameplay.
- **CPU (Central Processing Unit):** Upgrade to a faster processor to handle game calculations and prevent lag.
- **RAM (Random Access Memory):** Increase RAM to at least 16 GB for smooth gameplay and better multitasking.
- **SSD (Solid State Drive):** Install an SSD to reduce game loading times and make the system start faster.
- **Power Supply (PSU):** Upgrade to a suitable power supply to provide enough power for the new components.
- **Cooling System:** Improve cooling with better fans or a CPU cooler to prevent overheating during long gaming sessions.